

Algebra & Equations



Solving linear equations

- 1 Solve $3x - 7 = 11$.
 - 2 Solve $5(x - 2) = 3x + 4$.
 - 3 Solve $\frac{3x - 2}{4} = x - 1$.
 - 4 Solve $2(x + 3) - 5 = 3(x - 1)$.
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Expanding and simplifying expressions

- 5 Expand and simplify $3(2x - 1) - 2(x - 4)$.
 - 6 Expand $(x + 5)(x - 2)$.
 - 7 Expand $(2x - 3)^2$.
 - 8 Simplify $\frac{6x^2 - 9x}{3x}$ (for $x \neq 0$).
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Factoring quadratic expressions

- 9 Factor $x^2 + 7x + 12$.
 - 10 Factor $x^2 - 2x - 15$.
 - 11 Factor $2x^2 + 5x - 3$.
 - 12 Factor $x^2 - 49$ using the difference of squares.
-

Solving quadratic equations by factoring

- 13 Solve $x^2 - 5x + 6 = 0$.
- 14 Solve $x^2 + 2x - 8 = 0$.
- 15 Solve $2x^2 - 5x - 3 = 0$.

16 Solve $x^2 - 9x + 20 = 0$.

The quadratic formula and the discriminant

17 Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve $x^2 + 3x + 1 = 0$, leaving the answer in exact form.

18 Find the discriminant of $2x^2 - 4x + 5 = 0$ and state how many real roots it has.

19 Find the discriminant of $x^2 - 6x + 9 = 0$ and state how many real roots it has.

20 Use the quadratic formula to solve $3x^2 - 2x - 1 = 0$.

Absolute value equations

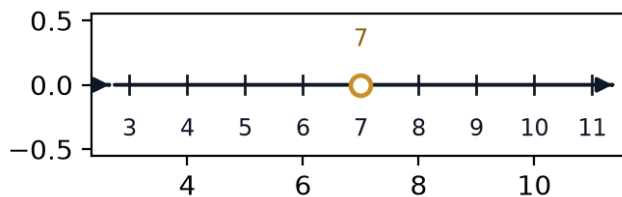
21 Solve $|x - 4| = 9$.

22 Solve $|2x + 3| = 7$.

23 Solve $3|x - 1| = 12$.

Linear inequalities

24 Solve $2x - 5 < 9$ and graph the solution set on a number line.



25 Solve $-3x + 4 \geq -8$.

26 Solve the compound inequality $5 < 2x + 1 \leq 11$.

Systems of linear equations

27 Solve the system $x + y = 10$, $x - y = 2$.

- 28 Solve the system $3x + 2y = 16$, $x - y = 2$.
- 29 This question has two parts. The sum of two numbers is 15 and their difference is 3.
- a Write a system of two equations for the numbers x and y .
 - b Solve the system to find both numbers.
-

Rational equations

- 30 Solve $\frac{x}{x-2} = 3$.
- 31 Solve $\frac{2}{x+1} = \frac{1}{3}$.
- 32 Solve $\frac{x+1}{x-1} = 2$.
-

Solving for a variable in a formula

- 33 The area of a triangle is $A = \frac{1}{2}bh$. Solve for h .
- 34 The circumference of a circle is $C = 2\pi r$. Solve for r .
- 35 The formula $v = v_0 + at$ gives velocity under constant acceleration. Solve for t .
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Linear-quadratic systems

- 36 Solve the system $y = x^2 - 1$, $y = 2x + 2$.
- 37 Solve the system $y = x^2$, $y = 3x - 2$.
-

Word problems: setting up and solving equations

- 38 This question has two parts. A rectangular garden has a length that is 4 meters more than its width w . Its area is 96 square meters.
- a Write an equation relating w to the area.
 - b Solve for w , and state the garden's length.

- 39 This question has two parts. A mobile-data plan charges a fixed monthly fee f plus a rate r per gigabyte used. One month, 5 GB of data cost a total of 65 dollars; another month, 9 GB cost a total of 93 dollars.
- a Write a system of two equations for f and r .
 - b Solve the system to find the fixed fee and the rate per gigabyte.
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MCQ — Solving linear equations

- 40 Solve $4x - 9 = 15$.

- a. $x = 6$
- b. $x = -6$
- c. $x = 24$
- d. $x = 1.5$

- 41 Solve $6(x - 3) = 2x + 10$.

- a. $x = -7$
- b. $x = 4$
- c. $x = 7$
- d. $x = 1$

- 42 Solve $\frac{2x - 1}{3} = x - 2$.

- a. $x = 5$
- b. $x = 1$
- c. $x = 7$
- d. $x = -5$

43 Solve $3(x + 4) - 7 = 2(x - 1)$.

a. $x = -7$

b. $x = -3$

c. $x = 3$

d. $x = 7$

44 Solve $5x + 3 = 2x - 9$.

a. $x = -2$

b. $x = 2$

c. $x = 4$

d. $x = -4$

MCQ — Expanding and simplifying expressions

45 Expand and simplify $4(3x - 2) - 3(x - 5)$.

a. $15x - 23$

b. $9x + 23$

c. $9x + 7$

d. $9x - 23$

46 Expand $(x + 7)(x - 3)$.

a. $x^2 + 10x - 21$

b. $x^2 - 4x - 21$

c. $x^2 + 4x - 21$

d. $x^2 + 4x + 21$

- 47 Expand $(3x - 1)^2$.
- a. $3x^2 - 6x + 1$
 - b. $9x^2 - 1$
 - c. $9x^2 - 6x + 1$
 - d. $9x^2 - 6x - 1$
- 48 Simplify $\frac{8x^2 - 12x}{4x}$ (for $x \neq 0$).
- a. $2x - 12$
 - b. $8x - 3$
 - c. $2x - 3$
 - d. $2x^2 - 3$

MCQ — Factoring quadratic expressions

- 49 Factor $x^2 + 9x + 14$.
- a. $(x + 2)(x + 7)$
 - b. $(x - 2)(x - 7)$
 - c. $(x + 1)(x + 14)$
 - d. $(x + 14)(x - 1)$
- 50 Factor $x^2 - 3x - 18$.
- a. $(x - 9)(x + 2)$
 - b. $(x - 2)(x + 9)$
 - c. $(x - 6)(x + 3)$
 - d. $(x + 6)(x - 3)$

51 Factor $3x^2 + 7x + 2$.

a. $(3x + 1)(x + 2)$

b. $(3x - 1)(x - 2)$

c. $(3x + 2)(x + 1)$

d. $(x + 1)(3x + 2)$

52 Factor $x^2 - 81$ using the difference of squares.

a. $(x - 9)^2$

b. $(x - 9)(x + 9)$

c. $(x + 9)^2$

d. $(x - 81)(x + 1)$

53 Factor $2x^2 - x - 6$.

a. $(2x + 3)(x - 2)$

b. $(2x - 3)(x + 2)$

c. $(x + 3)(2x - 2)$

d. $(2x + 1)(x - 6)$

MCQ — Solving quadratic equations by factoring

54 Solve $x^2 - 7x + 10 = 0$.

a. $x = 1$ or $x = 10$

b. $x = 7$ or $x = 10$

c. $x = -2$ or $x = -5$

d. $x = 2$ or $x = 5$

55 Solve $x^2 + 3x - 10 = 0$.

a. $x = -3$ or $x = 10$

b. $x = -10$ or $x = 1$

c. $x = 5$ or $x = -2$

d. $x = -5$ or $x = 2$

56 Solve $3x^2 - 11x - 4 = 0$.

a. $x = -\frac{1}{3}$ or $x = 4$

b. $x = \frac{1}{3}$ or $x = -4$

c. $x = 1$ or $x = 4$

d. $x = -\frac{1}{3}$ or $x = -4$

57 Solve $x^2 - 11x + 24 = 0$.

a. $x = 1$ or $x = 24$

b. $x = 4$ or $x = 6$

c. $x = -3$ or $x = -8$

d. $x = 3$ or $x = 8$

MCQ — The quadratic formula and the discriminant

58 Use the quadratic formula to solve $x^2 + 5x + 2 = 0$, leaving the answer in surd form.

a. $x = \frac{5 \pm \sqrt{17}}{2}$

b. $x = \frac{-5 \pm \sqrt{17}}{4}$

c. $x = \frac{-5 \pm \sqrt{33}}{2}$

d. $x = \frac{-5 \pm \sqrt{17}}{2}$

- 59 Find the discriminant of $3x^2 - 6x + 5 = 0$ and state how many real roots it has.
- a. 96; two real roots
 - b. 24; two real roots
 - c. -24; no real roots
 - d. -24; one real root
- 60 Find the discriminant of $x^2 - 8x + 16 = 0$ and state how many real roots it has.
- a. 0; one real root
 - b. 64; two real roots
 - c. 0; two real roots
 - d. -64; no real roots
- 61 Use the quadratic formula to solve $2x^2 + 3x - 2 = 0$.
- a. $x = 1$ or $x = -2$
 - b. $x = -0.5$ or $x = 2$
 - c. $x = 0.5$ or $x = -2$
 - d. $x = 0.5$ or $x = 2$
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MCQ — Absolute value equations

- 62 Solve $|x - 6| = 11$.
- a. $x = -17$ or $x = 5$
 - b. $x = 5$ or $x = -17$
 - c. $x = 17$ or $x = -5$
 - d. $x = 17$ or $x = 5$

63 Solve $|3x + 2| = 13$.

a. $x = \frac{11}{3}$ or $x = 5$

b. $x = 5$ or $x = -5$

c. $x = -\frac{11}{3}$ or $x = 5$

d. $x = \frac{11}{3}$ or $x = -5$

64 Solve $4|x - 2| = 20$.

a. $x = 7$ or $x = -3$

b. $x = -7$ or $x = 3$

c. $x = 5$ or $x = -5$

d. $x = 7$ or $x = 3$

65 Solve $|2x - 5| = 9$.

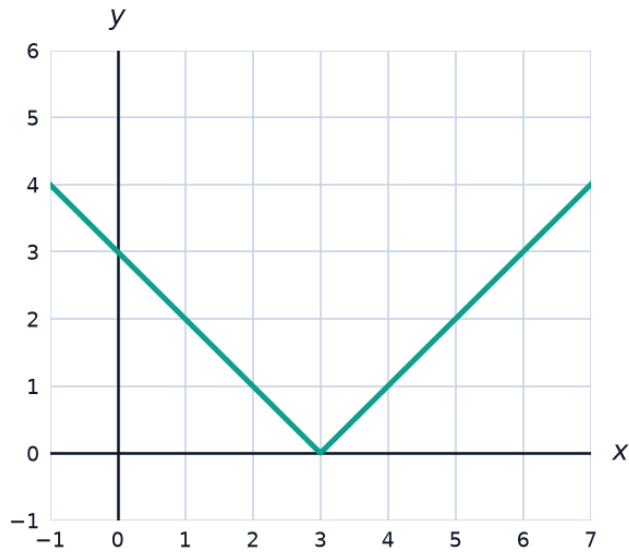
a. $x = 7$ or $x = 2$

b. $x = -7$ or $x = 2$

c. $x = 2$ or $x = -7$

d. $x = 7$ or $x = -2$

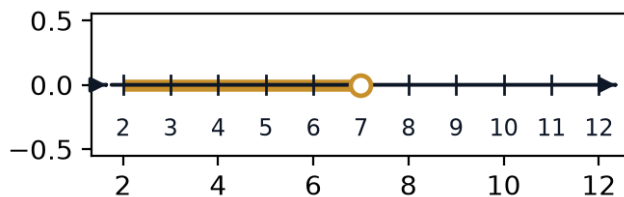
- 66 The graph shows $y = |x - 3|$. At what x -value does the graph reach its minimum?



- a. $x = 1$
- b. $x = 3$
- c. $x = -3$
- d. $x = 0$

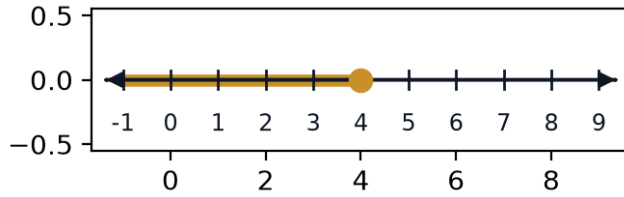
MCQ — Linear inequalities

- 67 Solve $3x - 7 < 14$, and identify the graph of its solution set.



- a. $x < 7$
- b. $x \geq 7$
- c. $x \leq 7$
- d. $x > 7$

- 68 Solve $-4x + 6 \geq -10$, and identify the graph of its solution set.



- a. $x > 4$
 - b. $x < 4$
 - c. $x \leq 4$
 - d. $x \geq 4$
- 69 Solve the compound inequality $7 < 3x + 1 \leq 19$.
- a. $2 < x \leq 6$
 - b. $2 < x < 6$
 - c. $2 \leq x < 6$
 - d. $6 < x \leq 2$
- 70 Solve $2x + 5 > -3$ and $x - 1 < 6$ together (both must hold). Find the combined solution set.
- a. $x > -4$
 - b. $-4 < x \leq 7$
 - c. $x < 7$
 - d. $-4 < x < 7$

MCQ — Systems of linear equations

71 Solve the system $x + y = 14$, $x - y = 4$.

a. $x = 5, y = 9$

b. $x = 10, y = 4$

c. $x = 9, y = 5$

d. $x = 9, y = 4$

72 Solve the system $4x + 3y = 26$, $x - y = 3$.

a. $x = 5, y = 2$

b. $x = 2, y = 5$

c. $x = 8, y = 5$

d. $x = 5, y = 3$

73 The sum of two numbers is 22 and their difference is 6. Find the two numbers.

a. 11 and 11

b. 14 and 6

c. 16 and 6

d. 14 and 8

74 Solve the system $2x - y = 7$, $3x + 2y = 21$.

a. $x = 5, y = -3$

b. $x = 5, y = 3$

c. $x = 3, y = 5$

d. $x = 7, y = 7$

MCQ — Rational equations

75 Solve $\frac{x}{x-3} = 4$.

a. $x = 12$

b. $x = -4$

c. $x = 4$

d. $x = 3$

76 Solve $\frac{3}{x+2} = \frac{1}{4}$.

a. $x = 6$

b. $x = 14$

c. $x = 10$

d. $x = -10$

77 Solve $\frac{x+2}{x-2} = 3$.

a. $x = 2$

b. $x = 8$

c. $x = 4$

d. $x = -4$

78 Solve $\frac{4}{x-1} = \frac{2}{x+2}$.

a. $x = -5$

b. $x = 5$

c. $x = -1$

d. $x = 2$

79 Solve $\frac{6}{x} = \frac{2}{x-4}$.

a. $x = 12$

b. $x = 4$

c. $x = -6$

d. $x = 6$

MCQ — Solving for a variable in a formula

80 The perimeter of a rectangle is $P = 2l + 2w$. Solve for l .

a. $l = \frac{P}{2} - w$

b. $l = \frac{P - 2w}{2}$

c. $l = \frac{P + 2w}{2}$

d. $l = P - 2w$

81 The formula for simple interest is $I = Prt$. Solve for r .

a. $r = \frac{I}{P}$

b. $r = \frac{I}{Pt}$

c. $r = \frac{Pt}{I}$

d. $r = I - Pt$

82 The formula $F = \frac{9}{5}C + 32$ converts Celsius to Fahrenheit. Solve for C .

a. $C = \frac{5(F - 32)}{9}$

b. $C = \frac{5F - 32}{9}$

c. $C = \frac{5}{9}F + 32$

d. $C = \frac{9(F - 32)}{5}$

83 The volume of a cylinder is $V = \pi r^2 h$. Solve for h .

a. $h = \frac{V}{\pi r^2}$

b. $h = V - \pi r^2$

c. $h = \frac{V}{\pi r}$

d. $h = \frac{\pi r^2}{V}$

MCQ — Linear-quadratic systems

84 Solve the system $y = x^2 - 4$, $y = 3x$.

a. $(-4, -12)$ and $(1, 3)$

b. $(-1, -3)$ and $(1, 3)$

c. $(4, 12)$ and $(1, 3)$

d. $(4, 12)$ and $(-1, -3)$

85 Solve the system $y = x^2$, $y = 2x + 3$.

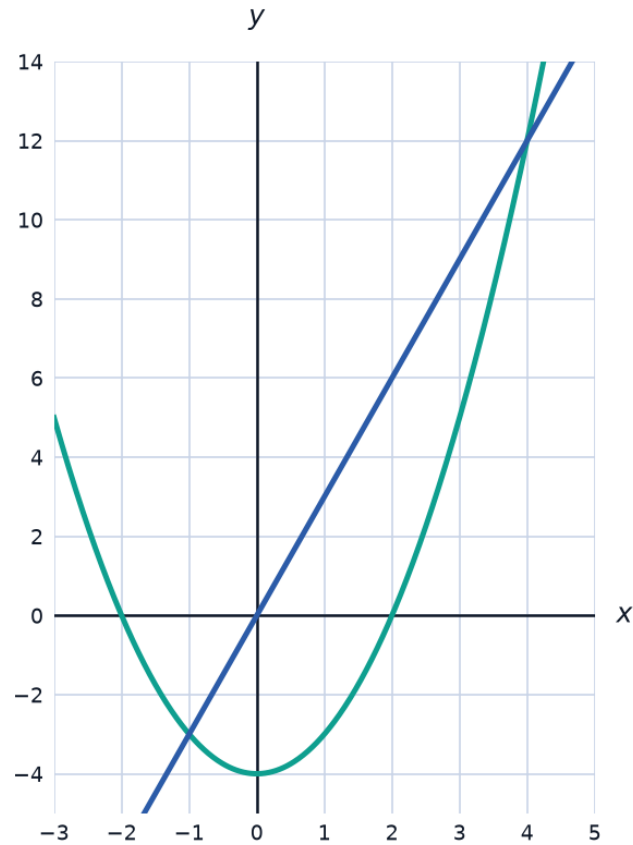
a. $(-3, 9)$ and $(1, 1)$

b. $(3, 9)$ and $(1, 1)$

c. $(-1, 1)$ and $(1, 1)$

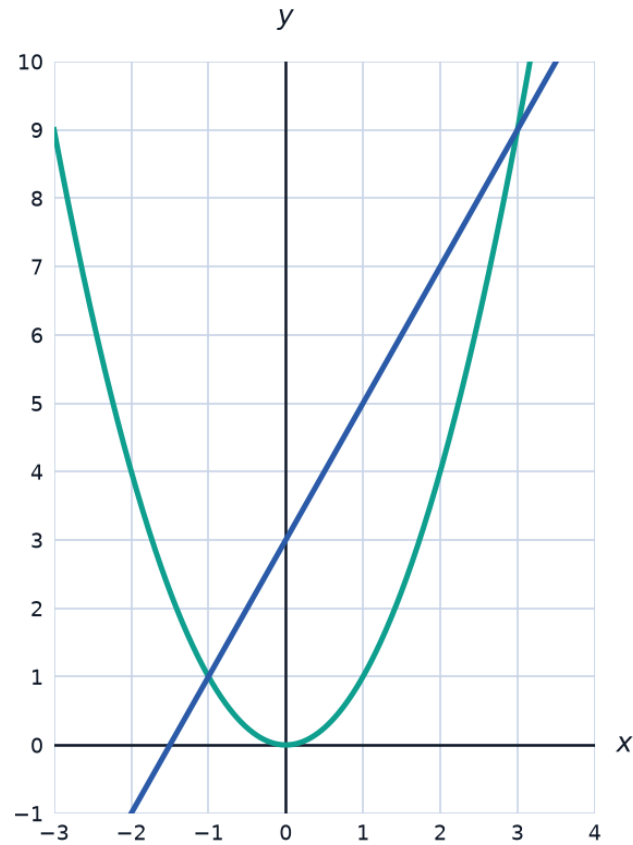
d. $(3, 9)$ and $(-1, 1)$

- 86 The graph shows the system $y = x^2 - 4$, $y = 3x$. What are the points of intersection?



- a. (4, 12) and (1, 3)
- b. (-4, -12) and (1, 3)
- c. (-1, -3) and (1, 3)
- d. (4, 12) and (-1, -3)

87 The graph shows the system $y = x^2$, $y = 2x + 3$. What are the points of intersection?



- a. (3, 9) and (1, 1)
- b. (-1, 1) and (1, 1)
- c. (-3, 9) and (1, 1)
- d. (3, 9) and (-1, 1)

MCQ — Word problems: setting up and solving equations

- 88 A rectangular garden has a length that is 5 meters more than its width, and its area is 84 square meters. Find the width.
- a. 12
 - b. 7
 - c. 9
 - d. 14
- 89 A mobile-data plan charges a fixed monthly fee of 40 riyals plus 2 riyals per GB used. If a user's bill was 68 riyals, how many GB did they use?
- a. 34
 - b. 14
 - c. 54
 - d. 28
-

Multiple choice — easy

- 90 Solve $4(x - 1) = x + 11$.
- a. -5
 - b. 3
 - c. 5
 - d. 7
- 91 Solve $4(x + 9) = x + 36$.
- a. 3
 - b. -2
 - c. 0
 - d. 2

92 Solve $4(x - 1) = 2x - 8$.

a. 2

b. 0

c. -4

d. -2

93 Solve $4(x + 7) = 3x + 25$.

a. 3

b. -1

c. -3

d. -5

94 Solve $5(x + 9) = x + 89$.

a. -11

b. 11

c. 9

d. 13

95 Solve $2(x + 7) = 3x + 23$.

a. -11

b. -7

c. -9

d. 9

96 Solve $5(x + 6) = x + 78$.

- a. 10
- b. 12
- c. -12
- d. 14

97 Solve $4(x - 9) = x - 12$.

- a. 10
- b. 8
- c. -8
- d. 6

98 Solve $5(x - 1) = x - 9$.

- a. 2
- b. -3
- c. 1
- d. -1

99 Solve $5(x + 6) = x + 18$.

- a. -1
- b. -5
- c. 3
- d. -3

100 Solve $5(x - 2) = 2x - 22$.

a. -2

b. -6

c. 4

d. -4

101 Solve $2(x - 1) = x + 7$.

a. 9

b. -9

c. 11

d. 7

102 Solve $2(x - 3) = 3x - 11$.

a. 3

b. 5

c. -5

d. 7

103 Solve $5(x + 9) = x + 57$.

a. 5

b. 1

c. 3

d. -3

104 Solve $2(x + 7) = x + 10$.

- a. 4
- b. -2
- c. -4
- d. -6

Multiple choice — medium

105 Factor $x^2 + 12x + 35$.

- a. $(x + 9)(x + 5)$
- b. $(x + 7)(x + 5)$
- c. $(x + 7)(x - 3)$
- d. $(x - 7)(x - 5)$

106 Factor $x^2 - 13x + 36$.

- a. $(x - 2)(x - 9)$
- b. $(x - 4)(x + 11)$
- c. $(x - 4)(x - 9)$
- d. $(x + 4)(x + 9)$

107 Factor $x^2 - 2x - 15$.

- a. $(x - 3)(x + 5)$
- b. $(x + 5)(x - 5)$
- c. $(x + 3)(x - 5)$
- d. $(x + 3)(x + 7)$

108 Factor $x^2 + 10x + 21$.

a. $(x + 7)(x - 1)$

b. $(x - 7)(x - 3)$

c. $(x + 7)(x + 3)$

d. $(x + 9)(x + 3)$

109 Factor $x^2 - x - 72$.

a. $(x - 9)(x - 6)$

b. $(x + 9)(x - 8)$

c. $(x - 9)(x + 8)$

d. $(x - 7)(x + 8)$

110 Factor $x^2 - 4x - 32$.

a. $(x - 6)(x + 4)$

b. $(x - 8)(x + 4)$

c. $(x + 8)(x - 4)$

d. $(x - 8)(x - 2)$

111 Factor $x^2 + 5x + 6$.

a. $(x - 3)(x - 2)$

b. $(x + 3)(x)$

c. $(x + 5)(x + 2)$

d. $(x + 3)(x + 2)$

112 Factor $x^2 + 3x - 28$.

a. $(x + 9)(x - 4)$

b. $(x + 7)(x + 6)$

c. $(x - 7)(x + 4)$

d. $(x + 7)(x - 4)$

113 Factor $x^2 - 8x + 15$.

a. $(x - 3)(x - 5)$

b. $(x + 3)(x + 5)$

c. $(x - 3)(x + 7)$

d. $(x - 1)(x - 5)$

114 Factor $x^2 + 5x - 14$.

a. $(x + 9)(x - 2)$

b. $(x + 7)(x + 4)$

c. $(x - 7)(x + 2)$

d. $(x + 7)(x - 2)$

115 Factor $x^2 + 4x - 32$.

a. $(x + 8)(x - 4)$

b. $(x + 10)(x - 4)$

c. $(x - 8)(x + 4)$

d. $(x + 8)(x + 6)$

116 Factor $x^2 + 11x + 18$.

a. $(x + 4)(x + 9)$

b. $(x + 2)(x - 7)$

c. $(x - 2)(x - 9)$

d. $(x + 2)(x + 9)$

117 Factor $x^2 + 6x - 16$.

a. $(x - 8)(x + 2)$

b. $(x + 10)(x - 2)$

c. $(x + 8)(x - 2)$

d. $(x + 8)(x + 4)$

118 Factor $x^2 - 36$.

a. $(x - 6)(x - 4)$

b. $(x - 4)(x + 6)$

c. $(x + 6)(x - 6)$

d. $(x - 6)(x + 6)$

119 Factor $x^2 - 9$.

a. $(x + 3)(x - 3)$

b. $(x - 3)(x + 3)$

c. $(x - 1)(x + 3)$

d. $(x - 3)(x - 1)$

Multiple choice — hard

120 Use the quadratic formula to solve $x^2 + x - 9 = 0$, leaving the answer in exact form.

- a. $\frac{1 \pm \sqrt{37}}{2}$
- b. $\frac{-1 \pm \sqrt{41}}{2}$
- c. $\frac{-1 \pm \sqrt{33}}{2}$
- d. $\frac{-1 \pm \sqrt{37}}{2}$

121 Use the quadratic formula to solve $2x^2 - 9x + 8 = 0$, leaving the answer in exact form.

- a. $\frac{-9 \pm \sqrt{17}}{4}$
- b. $\frac{9 \pm \sqrt{13}}{4}$
- c. $\frac{9 \pm \sqrt{21}}{4}$
- d. $\frac{9 \pm \sqrt{17}}{4}$

122 Use the quadratic formula to solve $x^2 - 8x - 5 = 0$, leaving the answer in exact form.

- a. $\frac{-8 \pm \sqrt{84}}{2}$
- b. $\frac{8 \pm \sqrt{84}}{2}$
- c. $\frac{8 \pm \sqrt{88}}{2}$
- d. $\frac{8 \pm \sqrt{80}}{2}$

123 Use the quadratic formula to solve $2x^2 + 5x - 8 = 0$, leaving the answer in exact form.

- a. $\frac{-5 \pm \sqrt{93}}{4}$
- b. $\frac{-5 \pm \sqrt{85}}{4}$
- c. $\frac{-5 \pm \sqrt{89}}{4}$
- d. $\frac{5 \pm \sqrt{89}}{4}$

124 Use the quadratic formula to solve $x^2 + 8x + 3 = 0$, leaving the answer in exact form.

- a. $\frac{-8 \pm \sqrt{56}}{2}$
- b. $\frac{8 \pm \sqrt{52}}{2}$
- c. $\frac{-8 \pm \sqrt{52}}{2}$
- d. $\frac{-8 \pm \sqrt{48}}{2}$

125 Find the discriminant of $x^2 - 3 = 0$ and state how many real roots it has.

- a. 12, one repeated real root
- b. 12, no real roots
- c. 12, two distinct real roots
- d. -12, two distinct real roots

126 Find the discriminant of $2x^2 - 2x + 4 = 0$ and state how many real roots it has.

- a. -28, no real roots
- b. -28, one repeated real root
- c. 28, no real roots
- d. -28, two distinct real roots

127 Find the discriminant of $2x^2 - 6x + 5 = 0$ and state how many real roots it has.

- a. -4, two distinct real roots
- b. -4, one repeated real root
- c. -4, no real roots
- d. 4, no real roots

128 Find the discriminant of $3x^2 - 4x + 9 = 0$ and state how many real roots it has.

- a. 92, no real roots
- b. -92, no real roots
- c. -92, two distinct real roots
- d. -92, one repeated real root

129 Find the discriminant of $2x^2 - 8x - 2 = 0$ and state how many real roots it has.

- a. 80, no real roots
- b. 80, two distinct real roots
- c. 80, one repeated real root
- d. -80, two distinct real roots

130 Solve $|3x + 6| = 18$.

- a. 4, -6
- b. 4, -8
- c. -8, 4
- d. 6, -8

131 Solve $|3x| = 3$.

- a. -1, 1
- b. 1, -1
- c. 1, 1
- d. 3, -1

132 Solve $|x + 9| = 11$.

a. $-20, 2$

b. $4, -20$

c. $2, -20$

d. $2, -18$

133 Solve $|x - 9| = 11$.

a. $-2, 20$

b. $22, -2$

c. $20, 0$

d. $20, -2$

134 Solve $|x - 6| = 17$.

a. $-11, 23$

b. $23, -11$

c. $23, -9$

d. $25, -11$